

CULT OPEN PROJECTS 2026

Recommendation Systems

for Personalized Content Discovery

Netflix Prize Dataset · SVD & Collaborative Filtering · Interactive Dashboard
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100M+

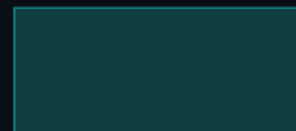
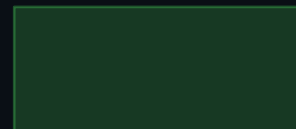
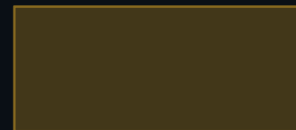
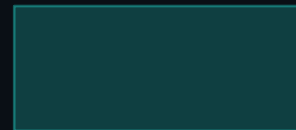
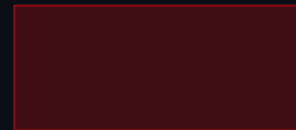
Ratings

480K

Users

17,770

Movies



Problem Overview

The Challenge

- 100M+ ratings across 480K users and 17,770 movies
- ~99% sparsity — users rate only a fraction of movies
- Long-tail activity — few power users dominate data
- Cold start problem for new/sparse users
- **Goal: deliver meaningful personalized recommendations**

Our Approach

01

EDA

Uncover patterns & sparsity

02

SVD Model

Matrix factorization baseline

03

CF Model

User-based similarity

04

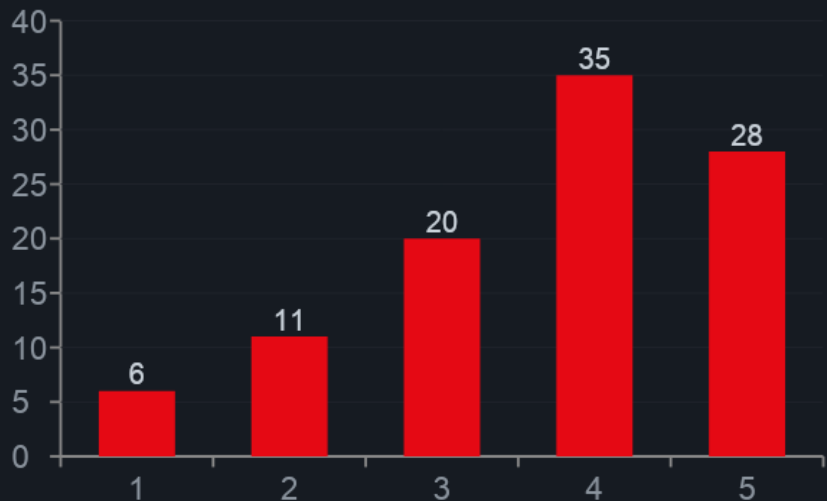
Dashboard

Live Streamlit deployment

Dataset permitted to be sampled — we used 500K ratings from combined_data_1.txt for reproducibility and computational efficiency

Exploratory Data Analysis

Rating Distribution



Key Findings



Left-Skewed

55% of ratings are 4–5 stars — selection bias (users rate what they watch)



Power Law Users

Median 12 ratings/user — sparse histories dominate the dataset



Long-Tail Movies

Few blockbusters attract most ratings; thousands of niche titles are rare



99% Sparsity

Matrix is nearly empty — demands latent factor generalization

Approach & Methodology



Models Implemented

Model 1 — SVD

Matrix Factorization

- 100 latent factors capturing hidden preferences
- 20 epochs · lr=0.005 · reg=0.02
- Decomposes $R \approx P \times Q^T$ across all users & movies
- Best baseline — mirrors Netflix Prize winning technique

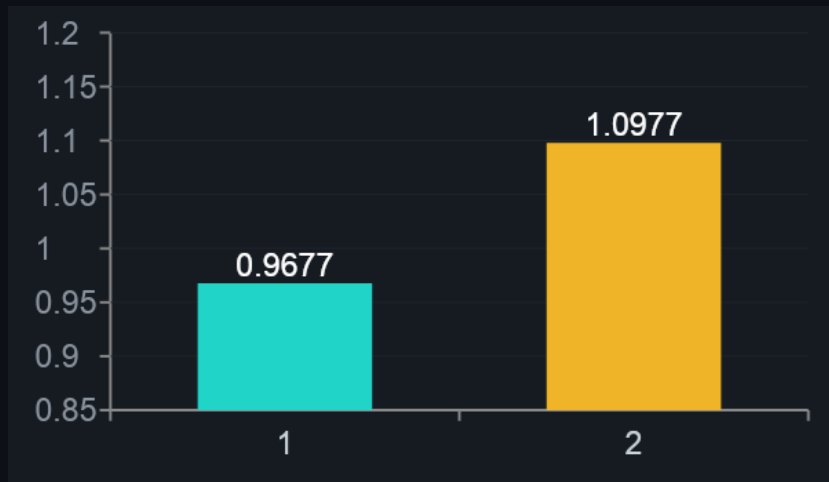
Model 2 — User-Based CF

K-Nearest Neighbours

- k=40 neighbors · cosine similarity · user-based
- Finds most similar users, recommends their liked movies
- Intuitive & explainable — 'users like you watched...'
- Struggles with sparsity — higher RMSE expected

Experimental Results

RMSE (lower = better)



MAP@10 (higher = better)



SVD wins on RMSE

0.9677 vs 1.0977 — SVD generalizes better in sparse data



Both strong MAP@10

85%+ relevant movies in top-10 for both models



Ranking > Prediction

MAP@10 is the business-relevant metric for users

Recommendation Examples

✓ Success Case — Active User (ID: 1933317)

Movies They Loved ★★★★★

- Star Trek: Voyager
- Justice League
- Lilo and Stitch
- Lord of the Rings
- The Game

SVD Top-5 Recommendations

- | | |
|---|------|
| 1. Lord of the Rings: Return of the King (Ext.) | 4.54 |
| 2. The Game | 4.42 |
| 3. Magnolia: Bonus Material | 4.40 |
| 4. Dragonheart | 4.35 |
| 5. ABC: Mel Gibson's The Passion of the Christ | 4.30 |

⚠ Failure Case — Sparse User (ID: 1982465) → Cold Start Problem

0 movies rated ≥ 4 ★ in sample → Predicted scores drop to 3.3–3.7 range → Less personalized, generic recommendations → Fix: popularity-based fallback for cold start users

Interactive Dashboard



Live: netflix-recommender-frv5acygcv6v9ffixmjtv1.streamlit.app



Home — EDA Dashboard

- Dataset stats (ratings, users, movies, sparsity)
- Rating distribution bar chart
- User activity histogram (log scale)
- Top 10 most-rated movies table



User Recommendations

- Enter any User ID → instant Top-10 recs
- Taste profile: avg rating, loved movies
- Confidence color coding (green/yellow/red)
- Explainability: why these recs were generated



Similar Movies

- Search any movie title
- Finds 10 similar movies via latent factor cosine similarity
- SVD qi vectors used — no extra training needed
- Serves as the Explainable AI component



Model Performance

- Side-by-side RMSE & MAP@10 bar charts
- Full methodology documentation
- Train/test split details
- Metric definitions & trade-off analysis

Covers: Optional Task C (Interactive Dashboard) + Optional Task E (Deployment) + Optional Task A (Explainable AI via latent factor similarity)

Key Insights & Future Work

Key Insights



SVD dominates CF

RMSE 0.9677 vs 1.0977 — latent factors handle sparsity better than local similarity



Both rank well

MAP@10 above 0.84 for both — relevant content consistently reaches top-10



Ranking > Accuracy

MAP@10 is the production-relevant metric — users care about order, not exact star predictions



Cold start is real

Sparse users get weak recs — popularity fallback needed for new users

Future Improvements

01

Full Dataset Training

Scale to all 100M ratings using distributed compute

02

SVD++

Incorporate implicit feedback for better cold start handling

03

Neural CF (NCF)

Deep learning to capture non-linear user-item interactions

04

Hybrid System

Combine SVD with content features (genre, year, director)